Mathematics practice booklet: Grade 8 expectations

Grade 8 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 8 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Draft: to be checked by a teacher before use.

Section 1: Patterns, equations and algebra thinking

Standards practised: 8.EE.B.5, 8.EE.B.6, 8.EE.C.7, 8.EE.C.8, 8.F.A.1, 8.F.A.2, 8.F.A.3, 8.F.B.4

A worked example

Solve: 3(x - 4) = 2x + 5. Then check your answer.

1. Expand the bracket: 3x - 12 = 2x + 5.
2. Take 2x from both sides: x - 12 = 5.
3. Add 12 to both sides: x = 17.
4. Check: 3 x (17 - 4) = 39 and 2 x 17 + 5 = 39. Both sides match.

x = 17

Questions

1. Solve: 3x + 7 = 28

2. Solve: 5(x - 2) = 3x + 6

3. Solve: x/2 + 3 = 11

4. Does 2x + 3 = 2x + 5 have a solution? Explain.

5. Find the slope of the line through (1, 2) and (4, 11).

6. Write the equation of the line with slope 2 that crosses the y-axis at (0, -3).

7. A taxi in Dubai charges AED 12 to start and AED 3 for each km. Write the cost y for x km, then find the cost of 10 km.

8. Solve the system: y = 2x + 1 and y = x + 4

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9. Solve the system: x + y = 10 and x - y = 4

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10. Is {(1, 3), (2, 5), (2, 7)} a function? Explain.

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11. Function A is y = 4x + 2. Function B has the table: x = 0, 1, 2 and y = 5, 7, 9. Which function grows faster?

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12. Salem has AED 50 and saves AED 20 a week. Hamdan has AED 10 and saves AED 30 a week. After how many weeks do they have the same amount, and how much is it?

Section 2: Shapes, measuring and space

Standards practised: 8.G.A.1, 8.G.A.2, 8.G.A.3, 8.G.A.4, 8.G.A.5, 8.G.B.6, 8.G.B.7, 8.G.B.8, 8.G.C.9

A worked example

Find the distance between (-2, 1) and (4, 9).

1. Horizontal change: 4 - (-2) = 6. Vertical change: 9 - 1 = 8.
2. These are the two legs of a right triangle: 6 squared + 8 squared = 36 + 64 = 100.
3. The distance is the square root of 100, which is 10.

10 units

Questions

1. Translate the point (2, 3) by 4 units right and 5 units down. If a whole triangle were translated like this, would the image be congruent to the original triangle?

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2. Reflect the point (3, -4) across the x-axis, then reflect the original point across the y-axis.

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3. Rotate the point (2, 5) by 90 degrees anticlockwise about the origin.

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4. A triangle is dilated from the origin with scale factor 3. The point (2, 4) is on it and one side is 2 cm long. Where does the point go and how long is the side now? Is the image congruent or similar?

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5. Triangle ABC has angles 40 and 60 degrees. Triangle DEF has angles 60 and 80 degrees. Are the triangles similar? Explain.

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6. Two angles of a triangle are 62 degrees and 47 degrees. Find the third angle.

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7. Two parallel lines are cut by a transversal. One angle is 65 degrees. Find its alternate interior angle and the same-side interior angle.

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8. A ladder leans against a wall. Its foot is 3 m from the wall and it reaches 4 m up the wall. How long is the ladder?

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9. A triangle has sides 8 cm, 15 cm and 17 cm. Is it a right triangle? Show why.

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10. Find the distance between (2, 1) and (7, 13).

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11. A cylindrical water tank has radius 2 m and height 5 m. Find its volume in terms of pi, then use 3.14 for pi.

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12. A cone has radius 3 cm and height 4 cm. A sphere has radius 3 cm. Find each volume (use 3.14 for pi).

Section 3: Data, graphs and chance

Standards practised: 8.SP.A.1, 8.SP.A.2, 8.SP.A.3, 8.SP.A.4

A worked example

In Grade 6, 20 students were asked about lunch: 15 bring a lunch box and 5 buy lunch. In Grade 8, 30 students were asked: 15 bring a lunch box and 15 buy lunch. Which grade is more likely to bring a lunch box?

1. The groups are different sizes, so compare percentages, not counts.
2. Grade 6: 15 out of 20 = 75%.
3. Grade 8: 15 out of 30 = 50%.
4. 75% is more than 50%.

Grade 6 (75% compared with 50%)

Questions

1. A scatter plot shows that on hotter days a shop sells more cold drinks. Describe the association.

2. A scatter plot of the age of cars and their selling price slopes downward. Describe the association.

3. The points on a scatter plot are spread out with no pattern. What can you say?

4. Points: (1, 2), (2, 4), (3, 6), (4, 8), (5, 20). Which point is an outlier?

5. A line of best fit for the walking time y (minutes) over x km is $y = 12x + 3$. Predict the time for 5 km.

6. In $y = 12x + 3$, what does the slope 12 mean in this situation?

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7. A line of best fit passes through (0, 4) and (10, 24). Write its equation.

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8. A survey of 50 students: Grade 7: 18 have a library card and 12 do not. Grade 8: 14 have a card and 6 do not. How many students have a card in all?

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9. In the same survey, what percentage of Grade 8 students have a card?

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10. What percentage of Grade 7 students have a card? Compare with Grade 8.

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11. Why is it better to compare percentages than counts in that survey?

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12. Pool visits and ice cream sales both rise in hot weather. Does a strong association mean ice cream causes more swimming?

Section 4: Numbers and calculating

Standards practised: 8.NS.A.1, 8.NS.A.2, 8.EE.A.1, 8.EE.A.2, 8.EE.A.3, 8.EE.A.4

A worked example

Write 0.666... (6 repeating) as a fraction. Then simplify 2^5 x 2^-2.

1. Let x = 0.666... Then 10x = 6.666...
2. Subtract: 10x - x = 6, so 9x = 6 and x = 6/9 = 2/3.
3. Same base, so add the exponents: 2^5 x 2^-2 = 2^3 = 8.

2/3 and 8

Questions

1. Write 5/8 as a decimal.

2. Write 0.272727... (27 repeating) as a fraction.

3. Which of these are irrational: the square root of 16, the square root of 20, pi, 0.5?

4. Between which two whole numbers does the square root of 50 lie?

5. Estimate the square root of 50 to the nearest tenth.

6. Which is bigger: the square root of 10 or 3.2?

7. Simplify and evaluate: 3^4 x 3^2

8. Simplify and evaluate: 5^7 / 5^4

9. Evaluate: (2^3)^2

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10. Evaluate: 2^-3 and 10^0

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11. The Hope Probe travelled about 493,000,000 km to reach Mars. Write this in scientific notation. Then write 0.00032 in scientific notation.

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12. How many times as large is 8 x 10^6 as 4 x 10^3? Write the answer in scientific notation.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. x = 7 (3x = 21)
2. x = 8 (5x - 10 = 3x + 6, so 2x = 16)
3. x = 16 (x/2 = 8)
4. No solution: taking 2x from both sides gives 3 = 5, which is never true.
5. 3 (rise 9 over run 3)
6. y = 2x - 3
7. y = 3x + 12; 10 km costs AED 42.
8. x = 3, y = 7 (2x + 1 = x + 4 gives x = 3, then y = 7)
9. x = 7, y = 3 (adding gives 2x = 14)
10. No: the input 2 has two different outputs, 5 and 7.
11. Function A: its rate of change is 4, while B's is 2.
12. After 4 weeks, AED 130 each (50 + 20w = 10 + 30w gives w = 4, and 50 + 20 x 4 = 130).

Section 2: Shapes, measuring and space

1. (6, -2). Yes, a translation keeps size and shape.
2. Across the x-axis: (3, 4). Across the y-axis: (-3, -4).
3. (-5, 2)
4. The point goes to (6, 12); the side is 6 cm. The image is similar, not congruent.
5. Yes. ABC's third angle is 80 and DEF's third angle is 40, so both have angles 40, 60, 80.
6. 71 degrees (the angles add to 180)
7. Alternate interior: 65 degrees. Same-side interior: 115 degrees (180 - 65).
8. 5 m (9 + 16 = 25)
9. Yes: 8 squared + 15 squared = 64 + 225 = 289 = 17 squared.
10. 13 units (changes of 5 and 12, and 25 + 144 = 169)
11. 20 pi cubic m, about 62.8 cubic m (pi x 4 x 5)
12. Cone: 12 pi, about 37.7 cubic cm. Sphere: 36 pi, about 113 cubic cm.

Section 3: Data, graphs and chance

1. Positive association: as one value goes up, so does the other.
2. Negative association: older cars tend to sell for less.
3. There is no association between the two variables.
4. (5, 20): it is far from the pattern of the others.
5. 63 minutes (12 x 5 + 3)
6. Each extra km adds about 12 minutes of walking.
7. y = 2x + 4 (slope 20 / 10 = 2)
8. 32 (18 + 14)
9. 70% (14 out of 20)
10. 60% (18 out of 30). A greater share of Grade 8 have a card, which suggests a small association.

11. The grades have different numbers of students (30 and 20), so counts would not be a fair comparison.
12. No: the heat explains both. Association does not prove one thing causes the other.

Section 4: Numbers and calculating

1. 0.625
2. 3/11 (27/99 simplified)
3. The square root of 20 and pi. (The square root of 16 is 4, and 0.5 is 1/2.)
4. Between 7 and 8 (49 < 50 < 64)
5. 7.1 (7.1 squared is 50.41)
6. 3.2 (3.2 squared is 10.24, which is more than 10)
7. 3^6 = 729
8. 5^3 = 125
9. 2^6 = 64
10. 1/8 and 1
11. 4.93 x 10^8 and 3.2 x 10^-4
12. 2 x 10^3 (8 / 4 = 2 and 10^6 / 10^3 = 10^3), so 2,000 times.